



APPLICATION FOR OBSERVING TIME

PERIOD: **95A**

Important Notice:

By submitting this proposal, the PI takes full responsibility for the content of the proposal, in particular with regard to the names of CoIs and the agreement to act according to the ESO policy and regulations, should observing time be granted.

1. Title										Category: D-5	
MUSE study of nearby core-collapse supernova host environments and parent stellar populations											
2. Abstract / Total Time Requested											
Total Amount of Time:											
We propose MUSE observations of nearby core-collapse supernova (SN) host galaxies, in order to study both the parent stellar population of the SN and the global host galaxy properties. The wide field of view and excellent spatial resolution of MUSE-IFU offers a unique opportunity to carry out this kind of study. The parent stellar population of the SN at the explosion site provides strong constraints on the age, initial mass, and metallicity of the once coeval SN progenitor stars, which are still largely unknown. This pilot study will test MUSE capabilities for future larger surveys aiming to connect SN and host galaxy/stellar population properties. Furthermore, the ability to resolve stellar populations across the host galaxy will enable comparison between the SN parent population and other stellar populations within the host galaxy as well as the galaxy as a whole, which can also be used to investigate possible biases arising from studying unresolved higher-redshift hosts.											
3. Run	Period	Instrument	Time	Month	Moon	Seeing	Sky	Mode	Type		
A	95	MUSE	3n	apr	d	0.8	THN	v			
4. Number of nights/hours											
a) already awarded to this project:				Telescope(s)				Amount of time			
b) still required to complete this project:				N/A				—			
5. Special remarks:											
This proposal is related to another P95 proposal (bad weather) to observe SN Ia host galaxies using MUSE (PI: Anderson). While either proposals have their own goals and are independent, the possible synergy and combination of datasets from both observations will contribute to a more complete picture of both core-collapse and thermonuclear SNe.											
6. Principal Investigator: hanin											
6a. Co-investigators:											
L.	Galbany	1823									
J.P.	Anderson	1261									
T.	Kruehler	1261									
M.	Hamuy	1823									
Following CoIs moved to the end of the document ...											

7. Description of the proposed programme

A – Scientific Rationale:

The progenitors of supernovae. Supernovae (SNe) are one of the most prominent transient events in astronomy. On its peak brightness, a SN may rival the light of the whole galaxy where it resides. SNe trigger new star formations and drive the cosmic chemical evolution. Despite their importance, the progenitors of SNe are still not very well characterized. In the context of stellar evolution, core-collapse SNe signal the death throes of stars. Massive stars around $\sim 8 M_{\odot}$ and above are believed to explode as type-II or Ib/c SNe. There have been numerous theoretical works modeling the evolution of massive stars up until the eventual SN (e.g. Heger et al. 2003, Groh et al. 2013), but the fundamental question still lingers: *is there any specific relation between the SN type and the characteristics of the progenitor star?*

During the last decade, there have been detections of SN progenitors in the pre-explosion archival Hubble Space Telescope images (see Smartt 2009 for a review). This is by far the most powerful way to directly constrain the SN progenitors; the photometry of the star gives its location in the H-R diagram. However, only a handful of SN progenitors have been directly detected since the method crucially depends on the availability of usable pre-explosion data. Alternatively, one may perform a statistical study to the environments of SNe to derive useful constraints on the progenitors. Among the SN subtypes, the type-IIP SN progenitors are the most well-characterized as red supergiant stars, while evidence is still scarce or non-existent for the other subtypes. In the case of direct progenitor imaging, the metallicity of the star is often assumed and extinction is still a major source of uncertainty.

Previous works. Using IFU spectroscopy to investigate SN parent stellar populations, Kuncarayakti et al. (2013a,b) have derived constraints on the mass and metallicity of 24 SNe of type-IIP/IIL/Ib/Ic. It was found that SN Ic progenitors are the most massive and metal-rich compared to the others, confirming previous studies such as Anderson et al. (2012) who proposed a progenitor mass sequence of SN Ic>Ib>II based on the association of the SNe to recent star formation, and also Modjaz et al. (2011) who found high metallicity for SN Ic. In a recent study using CALIFA IFU datasets, Galbany et al. (2014) found that SNe Ib/Ic occur closer to star-forming regions and higher star-forming density locations compared to the other SN types.

Kuncarayakti et al. (2013a) also proposed that there could be SN Ib/c progenitors which arise from binary systems from the stellar population age constraint. These binary progenitors lose their outer H/He envelope via close binary interaction to become SN IIL/Ib/c, and they do not necessarily reach $\gtrsim 25 M_{\odot}$ in order to remove the envelope via metallicity-driven stellar wind as is the case for single progenitor stars. As most massive stars are thought to undergo binary interaction (Sana et al. 2012) and some SN binary companions have been directly detected (e.g. Folatelli et al. 2014), the binarity issue is becoming increasingly important. If binarity is a dominant factor, metallicity may not be playing an important role and there should be no strong preference of SN types towards metallicity.

Scientific goal. We aim to obtain IFU spectroscopy of nearby SN sites presented in our target list, as well as the galaxy-wide environment of the SN. This will be made possible by the large MUSE field of view, which will enable us to collect IFU spectroscopy of the host galaxy in a small number of pointings. The parent stellar population of the SN located at the explosion site will provide the estimates for age, initial mass, and metallicity of the once coeval SN progenitor star. This will increase by 60% the number of published SN progenitors already constrained using this method. The IFU data of the SN host galaxy will be used to compare the SN parent population to other stellar populations within the galaxy in order to investigate whether the SN parent population is somewhat typical or probably there are different characteristics between different SN types. Extinction for the SN sites as well as across the whole host galaxy will also be determined, providing important insights on the characteristics of interstellar absorption towards SNe. Additionally, with the host galaxy resolved we will also be able to simulate it to be at higher redshift values and investigate the effects of aperture size, shape, and positioning, to the derived galaxy properties. This will be important for studies of SNe and galaxies at higher redshifts where the galaxies are not very well resolved and spectroscopy can only be done using much more limited techniques e.g. slit spectroscopy of the central part of the galaxy.

B – Immediate Objective:

Proposed observations & data analysis. We select core-collapse SNe residing in host galaxies within $cz < 1500 \text{ km s}^{-1}$, observable from the southern hemisphere ($\delta < 30^{\circ}$), having low inclination ($< 65^{\circ}$), and not too old to have reasonable astrometric accuracy (> 1980), which are not included yet in the catalog of nearby SN sites (Kuncarayakti et al. 2015, in prep.). The 13 SNe in 12 galaxies resulted from this selection are very well placed in RA for observations in April-May 2015. With MUSE, the explosion sites of these SNe will be observed as well as significant portions of the host galaxy.

Having worked with various IFU instruments and datasets, our team is very much experienced in the fields of integral field spectroscopy as well as in SN physics and galaxy/stellar population studies. Our experience in working with MUSE during the Science Verification phase would also provide great advantage that would jump-start the data reduction and basic analysis phase. Using MUSE datacubes from the observations proposed here we will be able to generate numerous maps (Fig.1, left) useful in characterizing host galaxy properties such as age, metallicity, and extinction maps. The age of the stellar population will be derived from spectral age

7. Description of the proposed programme and attachments

Description of the proposed programme (continued)

indicators (Fig.1, center) such as H α or Ca-triplet equivalent widths (compared to simple stellar population/SSP models such as Starburst99, Leitherer et al. 1999), metallicity via strong-line method (Pettini & Pagel 2004), and extinction from Balmer decrement. The initial mass of the SN progenitor would also be derived, using the age of the parent stellar population. This age equals the lifetime of the star, which was governed mainly by the initial mass. By comparing the lifetime to stellar evolution models, the initial mass of the SN progenitor star could be derived. Additionally, we will also compare the stellar population properties in different locations within the galaxy (Fig.1, right), as well as simulating the galaxy at different redshifts ($\sim$ resolutions). As higher-redshift SN hosts are often not resolved, this study will produce an important insight into possible biases that may arise from using unresolved host galaxies to characterize the embedded transients such as SNe and Gamma-Ray Bursts.

Expected outcome. With successful completion of the proposed observations, we will obtain datacubes of nearby core-collapse SN host galaxies encompassing the SN explosion sites, covering the whole optical wavelength range within 465–930nm with excellent spatial resolution of 0.2" ($\sim$ few tens of pc in linear scale). We will generate useful maps constraining physical properties of stellar populations present in the galaxy as well as the SN progenitor's parent population. The progenitor mass and metallicity will be derived, and the galaxy stellar populations will be characterized and studied to reveal patterns or correlations with the SN types and properties. Ultimately this will improve our knowledge of SN progenitors and their relation with the host galaxy stellar populations. This proposal will also open future opportunities for studying a larger sample of SN host galaxies using similar techniques.

Attachments (Figures)

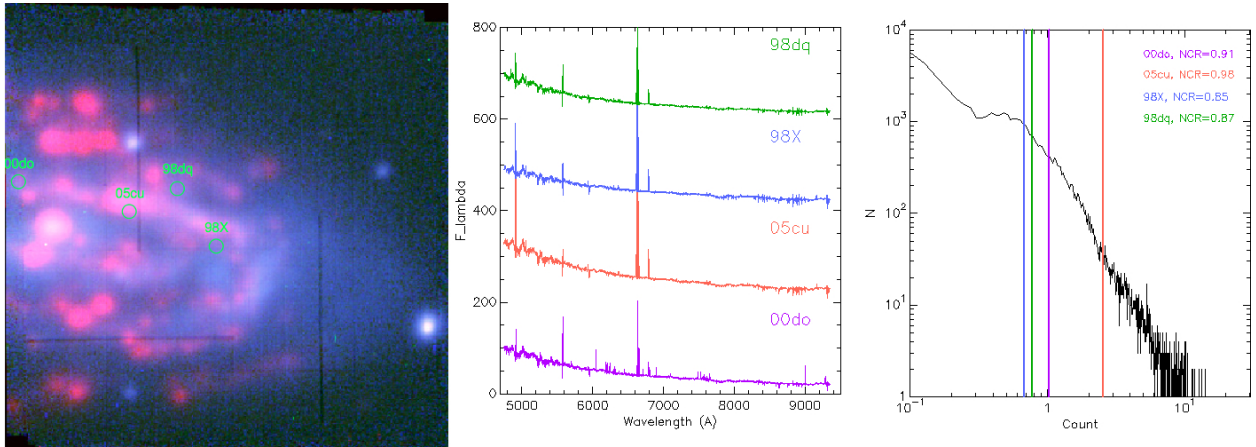


Fig. 1: An illustration of the expected observational data, from our observations during the MUSE Science Verification phase (Galbany et al. in prep.). (Left) An RGB composite image generated from one MUSE datacube, of the galaxy NGC6754 hosting four SNe. Red corresponds to H α emission wavelength, green 550nm and blue 480nm continuum wavelengths. The HII regions are readily identifiable (Center) The extracted spectra of the explosion sites of each SNe showing important emission lines useful for constraining the physical properties of the SN parent stellar population e.g. H α , [NII], H β , [OIII]. (Right) The H α emission strength at the SN explosion sites compared to the distribution of the whole galaxy H α pixels. With the proposed observation a large sample of this kind may be obtained, allowing for a more meaningful statistical study connecting SNe and the environments.

References can also be included using MakeCaption. For example:
References:

8. Justification of requested observing time and observing conditions

Lunar Phase Justification: Provide here a careful justification of the requested lunar phase.

Time Justification: (including seeing overhead) Provide a careful justification of the requested number of nights or hours for each observing run here. ESO Exposure Time Calculators exist for all Paranal and La Silla instruments and are available at the following web address:

<http://www.eso.org/observing/etc> .

Links to exposure time calculators for APEX instrumentation can be found in Section 7 of the Call for Proposals.

2x900s exp = 43 mins obs time, +sky: 3x900s = 1h

8a. Telescope Justification:

In order to carry out this kind of work, VLT/MUSE is the only instrument suitable. MUSE is currently the foremost instrument in the world capable of integral field spectroscopy. The fine spatial sampling of MUSE is needed to resolve individual stellar populations in the SN host galaxy, while the wide IFU field of view is important to capture significant part of the host galaxy.

8b. Observing Mode Justification (visitor or service):

We request visitor mode for observations in this proposal. Based on our experience, a visitor mode observation is more likely to be fully completed compared to service mode. Despite the fact that it is weather-dependent, the excellent site and facilities at Paranal Observatory almost guarantees that a visitor mode will be executed successfully. However, for flexibility we would also accept if our run is allocated for service mode.

8c. Calibration Request:

Standard Calibration

9. Report on the use of ESO facilities during the last 2 years

Proposal ID: 094.D-0290, 'Optical/NIR integral field spectroscopy of supernova explosion sites in nearby galaxies', PI: Kuncarayakti, VIMOS/SINFONI, VM+SM: This programme is still underway during the P94 observing semester, scheduled for observations in November 2014 and February 2015.

Proposal ID: 60.A-9344, 'Calibrating young simple stellar population (SSP) models with MUSE', PI: Kuncarayakti, MUSE: This programme was executed in MUSE Science Verification phase; the observation was shared with 2 other similar proposals (PIs: Vink, Fernandez). Data has been reduced and analysis is underway.

Proposal ID 60.A-9329, 'Characterising the location of SNe within galaxies', PI: Galbany, MUSE: Another MUSE SV proposal, the aim was to study SN host galaxies. Data reduced and analysis underway.

Proposal ID: 093.D-0318, 'Optical/NIR integral field spectroscopy of nearby supernova sites', PI: Kuncarayakti, VIMOS/SINFONI, VM: This programme was executed during the P93 semester with 100% of the observations completed. Data reduction has been completed, analysis is underway and a paper is in preparation.

9a. ESO Archive - Are the data requested by this proposal in the ESO Archive (<http://archive.eso.org>)? If so, explain the need for new data.

No.

9b. GTO/Public Survey Duplications:

N/A

10. Applicant's publications related to the subject of this application during the last 2 years

11. List of targets proposed in this programme

Run	Target/Field	α (J2000)	δ (J2000)	ToT	Mag.	Diam.	Additional info	Reference star
A	SN2009dq host	10 08 49.94	-67 01 57.3	0.75	V=13	1.5'	1-pointing	
A	SN2009md host	10 48 26.28	+12 13 02.7	1.5	g=13	2'	2-pointings	
A	SN2001fv host	11 04 02.988	+28 02 12.59	0.75	V=13	1'	1-pointing	
A	SN2007aa host	12 00 27.69	-01 04 51.6	2.25	g=13	4'	3-pointings	
A	SN2009N host	12 31 09.46	-08 02 56.3	2.25	R=12	4'	3-pointings	
A	SN2006my host	12 43 40.74	+16 23 14.1	1.5	V=11	3'	2-pointings	
A	SN1997X host	12 48 14.28	-03 19 58.5	1.5	V=11	2'	2-pointings	
A	SN1988E host	12 53 28.85	+02 10 22.6	2.25	V=11	4'	3-pointings	
A	SN2011jm host	12 54 51.10	+02 39 14.9	0.75	g=18	1'	1-pointing	
A	SN1999br host	13 00 41.80	+02 29 45.8	1.5	g=13	2'	2-pointings	
A	SN2007it host	14 18 25.63	-43 22 53.8	1.5	V=11	3'	2-pointings	
A	SN2001X host	15 21 55.45	+05 03 42.1	1.5	g=13	3'	2-pointings	

Target Notes: A note about the targets and/or strategy of selecting the targets during the run. For APEX runs please remember to specify the PWV limits for each target under 'Additional info' in the table above.

12. Scheduling requirements

13. Instrument configuration

Period	Instrument	Run ID	Parameter	Value or list
95	MUSE	A	WFM-NOAO-E	-

6b. Co-investigators:

...continued from Box 6a.

G.	Aldering	1463
N.	Arimoto	1010
M.	Doi	2050
Y.	Hashiba	2050
K.	Maeda	1436
T.	Morokuma	2050
T.	Usuda	1547